Lesson 6: Victory Condition

Scratch

VIDEO GAME CREATION

This week: a real win condition. Your job is to track how many enemies are left and show "You Win!" the moment the last one is destroyed - completing your game's Level 1 MVP milestone.

Open your own [YourName]Game project from Lesson 5 - don't start a new one. Everything this week builds on the row of enemies you already have.

Tip: no new assets this week - "You Win!" is shown with each engine's own basic text tools, not an image.

Setup

1. Open your [YourName]Game project from Lesson 5.
2. Create a new sprite (any simple placeholder costume), name it GameManager, and hide it at the green flag - it never needs to be seen, it just holds win-tracking logic.
3. Create a new variable enemies_remaining (for all sprites, not "for this sprite only").

Level 1 - Minimal (MVP)

The moment the last enemy is destroyed, "You Win!" appears on screen. This completes your game's Level 1 MVP milestone - a complete, playable core loop.

4. On GameManager, add "when green flag clicked" -> "hide" -> "set enemies_remaining to 5".
5. In the Enemy's "when I start as a clone" script, inside the existing "if touching Bullet? then" branch, replace the old decrement with "broadcast [enemy hit]" right before "delete this clone" - the Enemy now only reports that it was hit, it no longer counts anything itself.
6. On GameManager, add a new script: "when I receive [enemy hit]" -> "change enemies_remaining by -1" -> "if enemies_remaining = 0 then broadcast [you win]".
7. On the Player sprite, add a new script: "when I receive [you win]" -> "say [You Win!] for 3 seconds".
8. Click the stage, then shoot every enemy to test - the message should appear the instant the last one is destroyed.

Level 2 - Complete (Alpha/Beta)

The win text appears exactly once, exactly when the last enemy is destroyed - not early, not late, and not repeatedly.

- Destroy all 5 enemies and confirm "You Win!" appears exactly once, exactly when the last one is destroyed - not one enemy early, and not showing up multiple times.
- This already works correctly because all the counting logic lives in exactly one place - GameManager's "when I receive [enemy hit]" script - which runs once per broadcast, and each Enemy clone only ever broadcasts once, right when it's actually hit.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Sound: add "start sound" using the provided sfx_win.wav right before the "say" block on the Player's "when I receive [you win]" script.
- Visual: make the win message more of a celebration - try a few "change size by" pulses in a row, or a costume/color-effect change, instead of a plain "say" block.
- Structure: keep all win-tracking (enemies_remaining, the zero-check, the win broadcast) inside GameManager's own script - the Enemy sprite only ever reports that it was hit, it never counts or decides anything itself.

If Something's Wrong

- "You Win!" never appears: confirm GameManager's "when green flag clicked" script actually runs and sets enemies_remaining to 5 - an easy thing to miss since GameManager is hidden and easy to forget about.
- "You Win!" appears too early or too late: confirm enemies_remaining is being set to the correct starting count (matching your actual spawn loop's repeat count) on GameManager's green-flag script.
- Message appears multiple times: confirm GameManager has only ONE "when I receive [enemy hit]" script - a second copy would double-count every hit.

Want More?

Student Resource List > Scratch section, or scratch.mit.edu/ideas